Topic: Divergence Theorem

I. Introduction

To motivate this section, lets start with a problem

Compute the flux of $\mathbf{F} = [x^2, 0, z^2]$ through the rectangular box: $|x| \leq 1$, $|y| \leq 3$, and $0 \leq z \leq 2$. Let S be the surface of the box.

Note: Since S is composed of 6 surfaces, we need to compute the flux through each surface.

$$
\begin{aligned}
Flux &= \iint_S \mathbf{F} \cdot \mathbf{n}\, dA \\
&= \iint_{S_{x-}} \mathbf{F} \cdot \mathbf{n}_{x-}\, dA + \iint_{S_{x+}} \mathbf{F} \cdot \mathbf{n}_{x+}\, dA + \iint_{S_{y-}} \mathbf{F} \cdot \mathbf{n}_{y-}\, dA + \iint_{S_{y+}} \mathbf{F} \cdot \mathbf{n}_{y+}\, dA + \iint_{S_{z-}} \mathbf{F} \cdot \mathbf{n}_{z-}\, dA + \iint_{S_{z+}} \mathbf{F} \cdot \mathbf{n}_{z+}\, dA
\end{aligned}
$$

S_{x-}: $x =$, $\leq y$, $\leq z \leq$, and $\mathbf{n}_{x-} =$

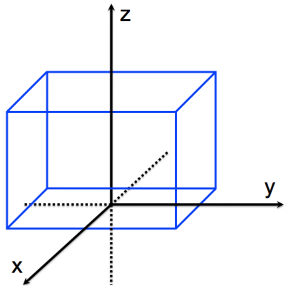
S_{x+}: $x =$, $\leq y \leq$, $\leq z \leq$, and $\mathbf{n}_{x+} =$

S_{y-}: $y =$, $\leq x \leq$, $\leq z \leq$, and $\mathbf{n}_{y-} =$

S_{y+}: $y =$, $\leq x \leq$, $\leq z \leq$, and $\mathbf{n}_{y+} =$

S_{z-}: $z =$, $\leq x \leq$, $\leq y \leq$, and $\mathbf{n}_{z-} =$

S_{z+}: $z =$, $\leq x \leq$, $\leq y \leq$, and $\mathbf{n}_{z+} =$



$$Flux = \iint\limits_{S_{x-}} \mathbf{F} \cdot \mathbf{n}_{x-}\, dA + \iint\limits_{S_{x+}} \mathbf{F} \cdot \mathbf{n}_{x+}\, dA + \iint\limits_{S_{y-}} \mathbf{F} \cdot \mathbf{n}_{y-}\, dA + \iint\limits_{S_{y+}} \mathbf{F} \cdot \mathbf{n}_{y+}\, dA + \iint\limits_{S_{z-}} \mathbf{F} \cdot \mathbf{n}_{z-}\, dA + \iint\limits_{S_{z+}} \mathbf{F} \cdot \mathbf{n}_{z+}\, dA$$

$$\iint\limits_{S_{x-}} \mathbf{F} \cdot \mathbf{n}_{x-}\, dA =$$

$$\iint\limits_{S_{x+}} \mathbf{F} \cdot \mathbf{n}_{x+}\, dA =$$

$$\iint\limits_{S_{y-}} \mathbf{F} \cdot \mathbf{n}_{y-}\, dA =$$

$$\iint\limits_{S_{y+}} \mathbf{F} \cdot \mathbf{n}_{y+}\, dA =$$

$$\iint\limits_{S_{z-}} \mathbf{F} \cdot \mathbf{n}_{z-}\, dA =$$

$$\iint\limits_{S_{z+}} \mathbf{F} \cdot \mathbf{n}_{z+}\, dA =$$

$$\implies \quad Flux =$$

Fortunately, just like Green's Theorem, there is a relationship between the boundary and the enclosed solid that makes this easier.

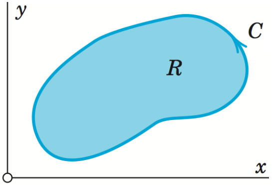
II. Divergence Theorem

Recall: Green's Thm relates a with the

The Divergence Theorem relates a with the

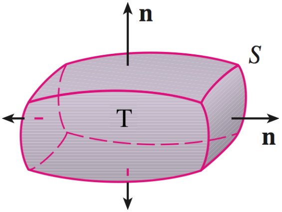
<u>Thm</u>: Let $\mathbf{F} = [M(x,y,z), N(x,y,z), P(x,y,z)]$ be a vector field that has continuous 1st partial derivatives. Also, let T be a bounded volume with boundary S (oriented outward) which is imbedded in $\mathbf{F}$. Then

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dA =$$



Green's Theorem

$$\int_C \mathbf{F} \cdot d\mathbf{r} =$$



Divergence Theorem

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dA =$$

Back to our first problem:

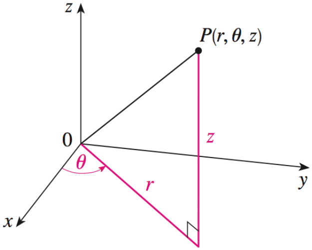
Ex: Compute the flux of $\mathbf{F} = [x^2, 0, z^2]$ through the rectangular box: $|x| \leq 1$, $|y| \leq 3$, and $0 \leq z \leq 2$.

$$Flux =$$

III. Review: Cylindrical & Spherical Coordinates

Since we will be computing triple integrals, we should review the different types of coordinates that we may use.

Cylindrical Coordinates:



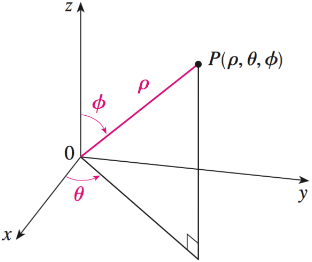
$x =$

$y =$

$z =$

-
- $dV =$

Spherical Coordinates:



$x =$

$y =$

$z =$

-
- $dV =$

Ex: Evaluate $\iint_S \mathbf{F}\cdot\mathbf{n}\,dA$ if $F = [x^3 - y^3, y^3 - z^3, z^3 - x^3]$ and S is given by: $x^2 + y^2 + z^2 \le 25,\ \ z \ge 0$

$$\iint_S \mathbf{F}\cdot\mathbf{n}\,dA =$$

Now we need to choose our coordinates. Since S is a sphere, we should use spherical coordinates.

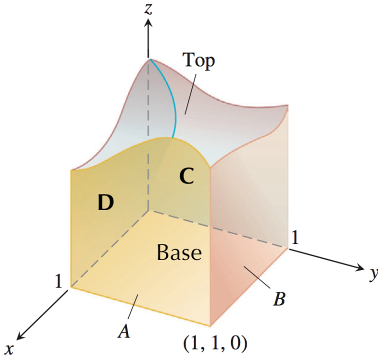
Remember:

Ex: The the closed cube-like surface, S, shown below has a top that is an unknown smooth surface. Let $\mathbf{F} = x\mathbf{i} - 2y\mathbf{j} + (z+3)\mathbf{k}$ and suppose the outward flux through side A is 1 and through side B is - 3. Find the outward flux through the top?

Let S be the total surface and T the solid bounded by S

Want:

We are given:



For the Base, we know: and

$\implies \mathbf{F} \cdot \mathbf{n} =$

$$\implies \iint\limits_{Base} \mathbf{F} \cdot \mathbf{n} \, dA =$$

For Side C, we know: and

$\implies \mathbf{F} \cdot \mathbf{n} =$

$$\implies \iint\limits_{C} \mathbf{F} \cdot \mathbf{n} \, dA =$$

For Side D, we know: and

$\implies \mathbf{F} \cdot \mathbf{n} =$

$$\implies \iint\limits_{D} \mathbf{F} \cdot \mathbf{n} \, dA =$$

Updating (*), we have:

Therefore,